

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN CLIMATE

MODULE - 2

INDIAN MONSOON

- **FACTORS DETERMINING THE CLIMATE OF INDIA**

FACTORS DETERMINING THE CLIMATE OF INDIA

India's climate is controlled by a number of factors which can be broadly divided into two groups:

1. Factors related to **location and relief**
2. Factors related to **air pressure and winds**.

POINT NO:2

FACTORS RELATED TO AIR PRESSURE & WINDS

These factors related to air pressure and winds can be further sub-divided into the following:

1. Monsoon wind movement
2. Upper Air Circulation- Easterly and Westerly Jetstream
3. Tropical cyclones and Western Disturbances
4. El Nino effect
5. La Nina effect
6. Southern Oscillation

**EFFECT DURING
SUMMER AND
WINTER SEASON**

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO AIR PRESSURE & WINDS

Let us see these factors in much detail now...

- 1** To understand the differences in local climates of India, we need to understand the mechanism of the following three important factors :
 - i. **Distribution of air pressure and winds** on the surface of the earth also known as Monsoon Wind Movement (Sub-factor 1).
 - ii. **Upper air circulation** caused by factors controlling global weather and the inflow of different air masses and jet streams (Sub-factor 2).
 - iii. **Inflow of western cyclones** generally known as disturbances during the winter season and **tropical depressions** during the south-west monsoon period into India, creating weather conditions favourable to rainfall (Sub-factor 3).
- 2** The mechanism of these three factors can be understood with reference to **winter and summer seasons** of the year separately.

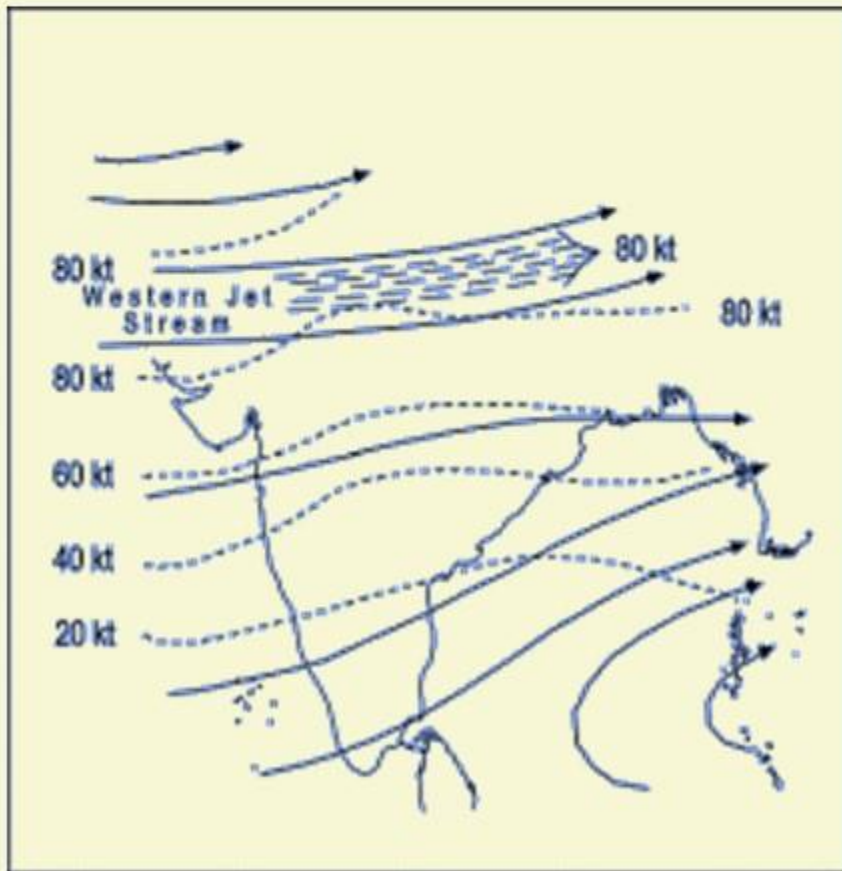
FACTORS RELATED TO AIR PRESSURE & WINDS

1

MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:1

SURFACE PRESSURE AND MONSOON WINDS



1

In winter months, the weather conditions over India are generally influenced by the distribution of pressure in Central and Western Asia.

2

A high pressure centre in the region lying to the north of the Himalayas develops during winter.

3

This centre of high pressure gives rise to the flow of air at the low level from the north towards the Indian subcontinent, south of the mountain range.

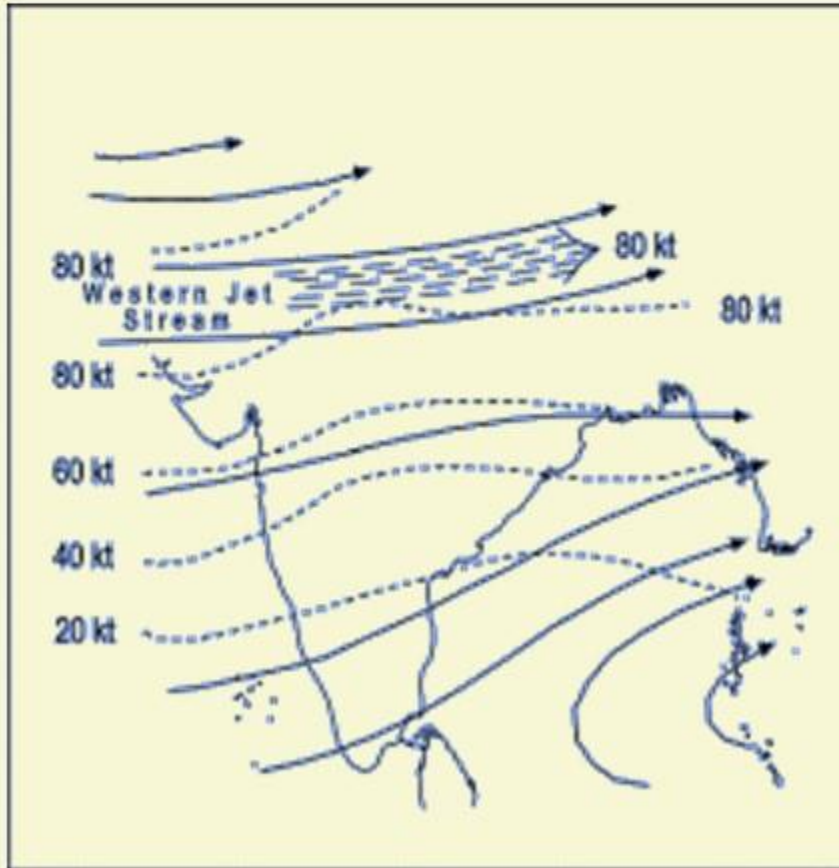
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MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:1

SURFACE PRESSURE AND MONSOON WINDS



4

The surface winds blowing out of the high pressure centre over Central Asia reach India in the form of a **dry continental air mass**.

5

These continental winds come in contact with **west blowing winds over north western India**.

6

The position of this contact zone is not, however, stable.

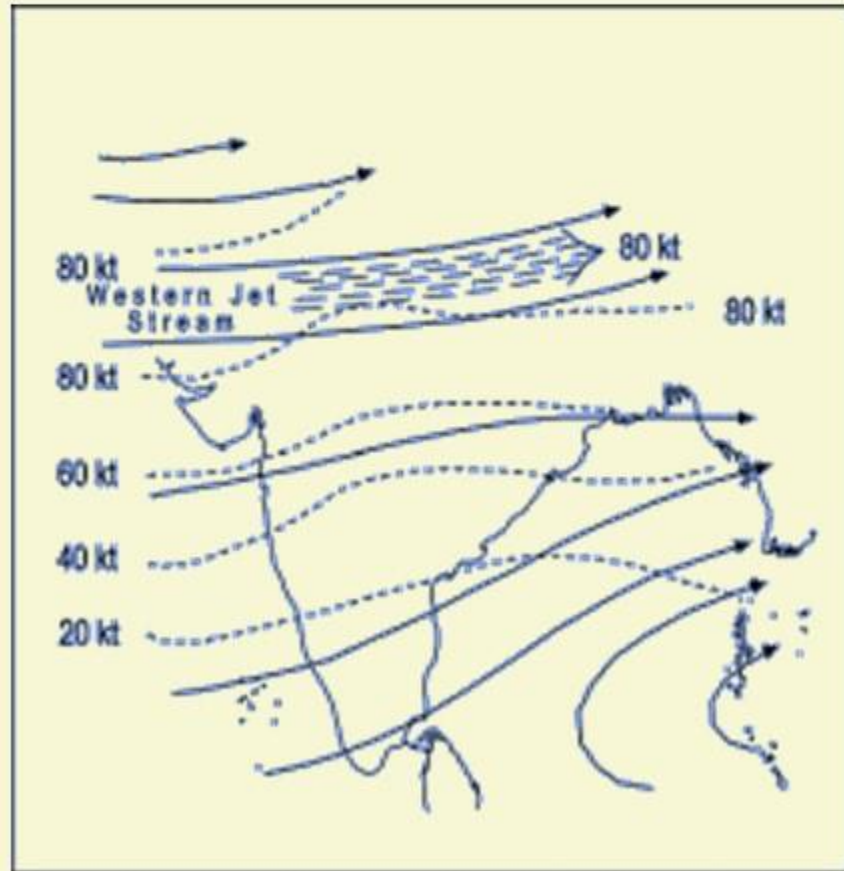
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MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:1

SURFACE PRESSURE AND MONSOON WINDS



7

Occasionally, it may shift its position as far east as the middle Ganga valley with the result that the whole of the northwestern and northern India up to the middle Ganga valley comes under the influence of dry northwestern winds.

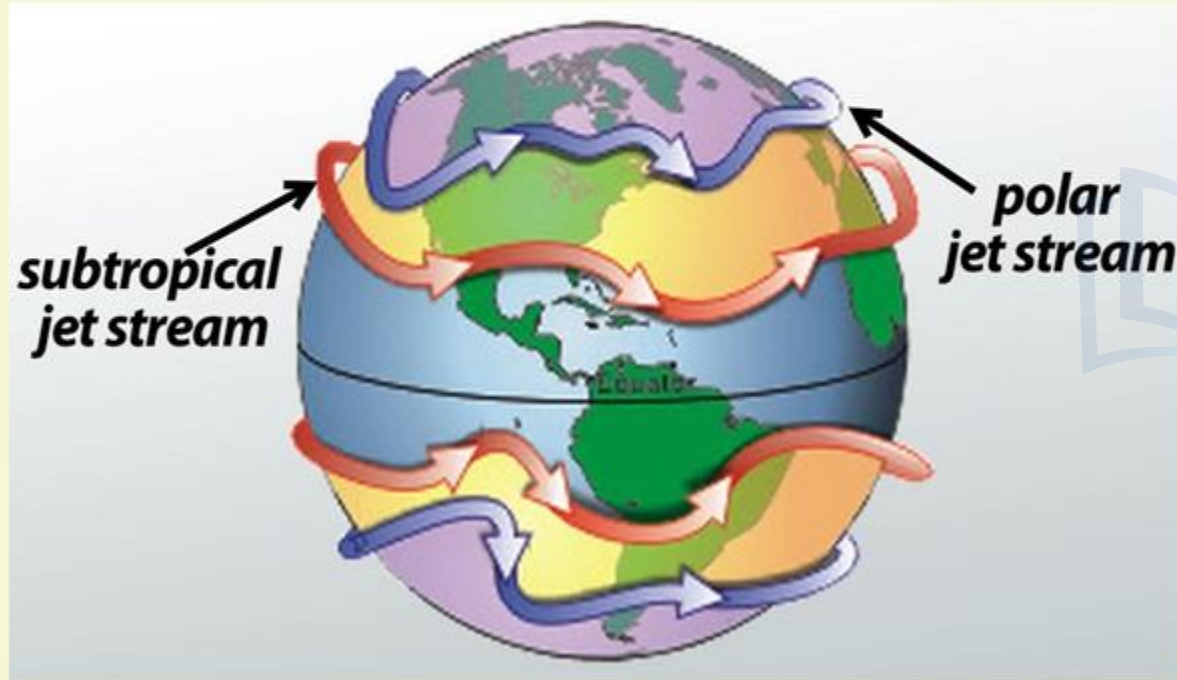
FACTORS RELATED TO AIR PRESSURE & WINDS

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MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:2

JET STREAM AND UPPER AIR CIRCULATION



1

The pattern of air circulation discussed above is witnessed **only at the lower level of the atmosphere** near the surface of the earth.

2

Higher up in the lower troposphere, a different pattern of air circulation is observed.

3

The variations in the atmospheric pressure closer to the surface of the earth have no role to play in the making of upper air circulation.

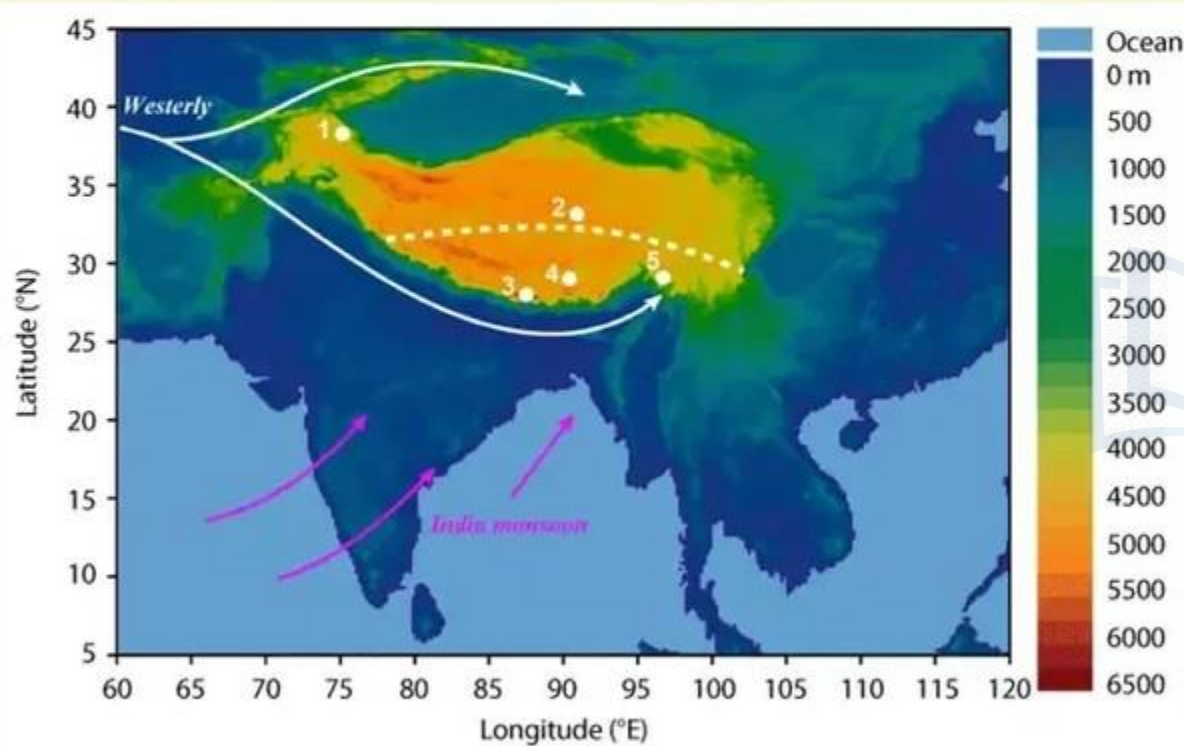
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MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:2

JET STREAM AND UPPER AIR CIRCULATION



Bifurcation of Jetstream by Tibetan Highlands

4

All of Western and Central Asia remains under the influence of westerly winds along the altitude of 9-13 km from west to east.

5

These winds blow across the Asian continent at latitudes north of the Himalayas roughly parallel to the Tibetan highlands. These are known as jet streams.

6

Tibetan highlands act as a barrier in the path of these jet streams. As a result, jet streams get bifurcated.

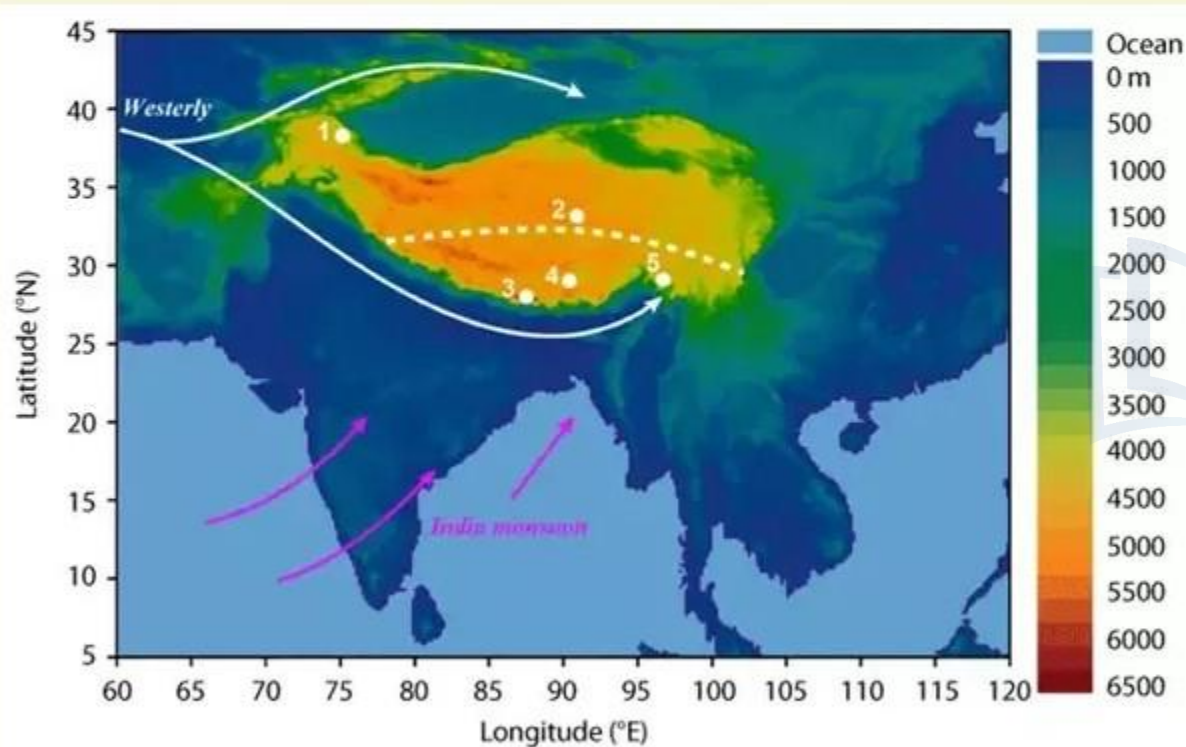
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Bifurcation of Jetstream by Tibetan Highlands

7

One of its branches blows to the north of the Tibetan highlands, while the southern branch blows in an eastward direction, south of the Himalayas.

8

It has its mean position at 25°N in February at 200-300 mb level.

9

It is believed that this **southern branch of the jet stream** exercises an important influence on the winter weather in India.

JET STREAM

TYPES OF JET STREAMS

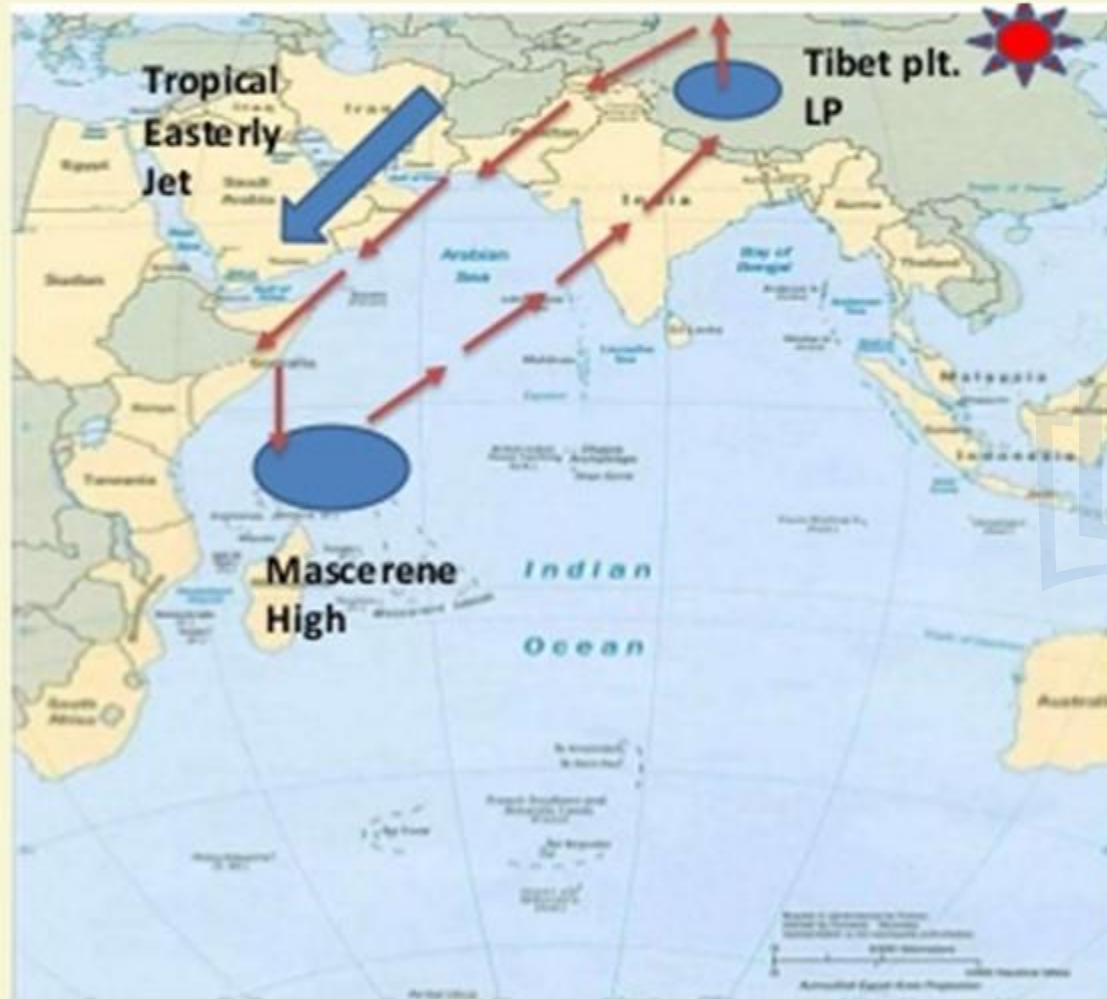


2. SUBTROPICAL WESTERLY JET STREAMS

- 1 Subtropical westerly jet streams move in the upper troposphere to the north of subtropical surface high pressure belt (at the poleward limit of the Hadley cell in both the hemispheres) i.e above 30° - 35° latitudes.
- 2 Their circulation is from the west to east in more regular manner than the polar front jet streams

JET STREAM

TYPES OF JET STREAMS



3. TROPICAL EASTERLY JET STREAMS

Tropical easterly jet streams develop in the upper troposphere **above surface easterly trade winds** over India and Africa during summer season due to intense heating of **Tibetan plateau** and play important role in the mechanism of Indian monsoon

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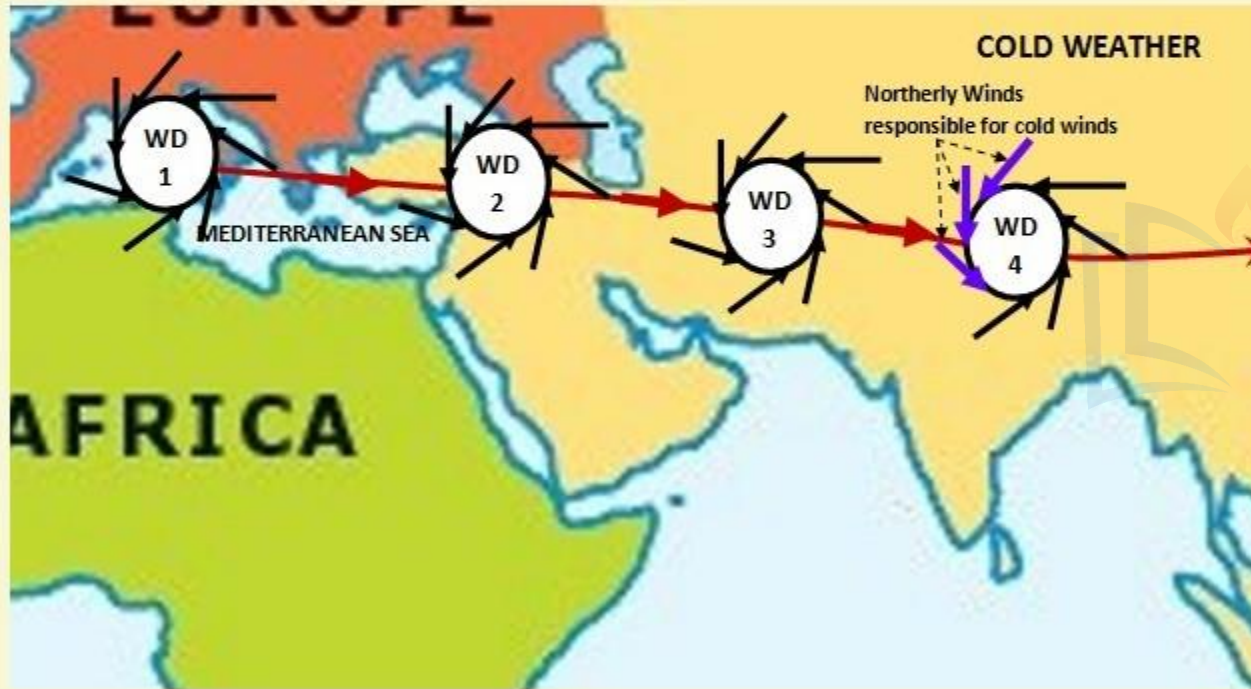
FACTORS RELATED TO AIR PRESSURE & WINDS

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MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:3

WESTERN CYCLONIC DISTURBANCE AND TROPICAL CYCLONES



1

The western cyclonic disturbances which enter the Indian subcontinent from the west and the northwest during the winter months, originate over the Mediterranean Sea and are brought into India by the westerly jet stream.

2

An increase in the prevailing night temperature generally indicates an advance in the arrival of these cyclones disturbances.

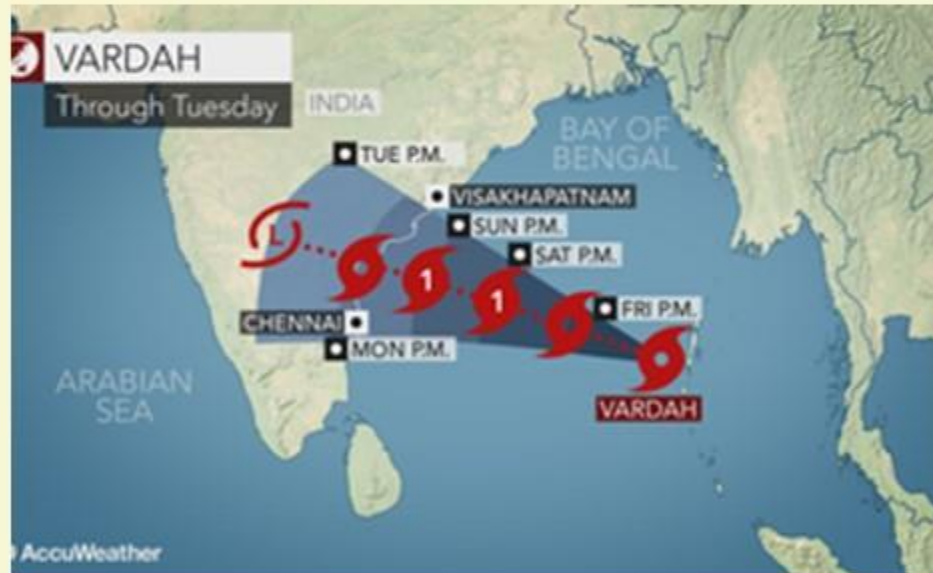
FACTORS RELATED TO AIR PRESSURE & WINDS

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MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:3

WESTERN CYCLONIC DISTURBANCE AND TROPICAL CYCLONES



3

Tropical cyclones originate over the Bay of Bengal and the Indian ocean.

4

These tropical cyclones have very high wind velocity and heavy rainfall and hit the Tamil Nadu, Andhra Pradesh and Orissa coast.

5

Most of these cyclones are very destructive due to high wind velocity and torrential rain that accompanies it.

6

Have you seen their movement in the weather report in the television?

FACTORS RELATED TO AIR PRESSURE & WINDS

1 MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:3

WESTERN CYCLONIC DISTURBANCE AND
TROPICAL CYCLONES

QUESTION

Assertion: The Kullu valley in Himachal Pradesh receives copious snowfall during winter.

Reason: The Kullu valley receives moisture bearing wind of the Western Disturbances during the winter.

- a) Both statements are true and statement 2 perfect explanation of statement 1.
- b) Both statements are true but 2 is not correct explanation of 1.
- c) Statement 1 is true and 2 is false
- d) None of the statements is true

FACTORS RELATED TO AIR PRESSURE & WINDS

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FACTORS RELATED TO AIR PRESSURE & WINDS

1 MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:3

WESTERN CYCLONIC DISTURBANCE AND TROPICAL CYCLONES

WHAT ARE WESTERN DISTURBANCES?

- During the winters, the weather in India is pleasant. The pleasant weather conditions, however, at intervals, get disturbed by shallow cyclonic depressions originating over the east Mediterranean Sea.
- These cyclones travel eastwards across West Asia, Iran, Afghanistan and Pakistan and enter the north-western parts of India with the help of the westerly jetstream.
- On their way, the moisture content gets augmented from the Caspian Sea in the north and the Persian Gulf in the south.
- Since these cyclones reach India from the West, they are usually referred to as the western disturbances over the Indian region.

FAST AND FURIOUS

The extreme weather conditions are a result of a weather phenomenon called western disturbance, which is going to be quite intense

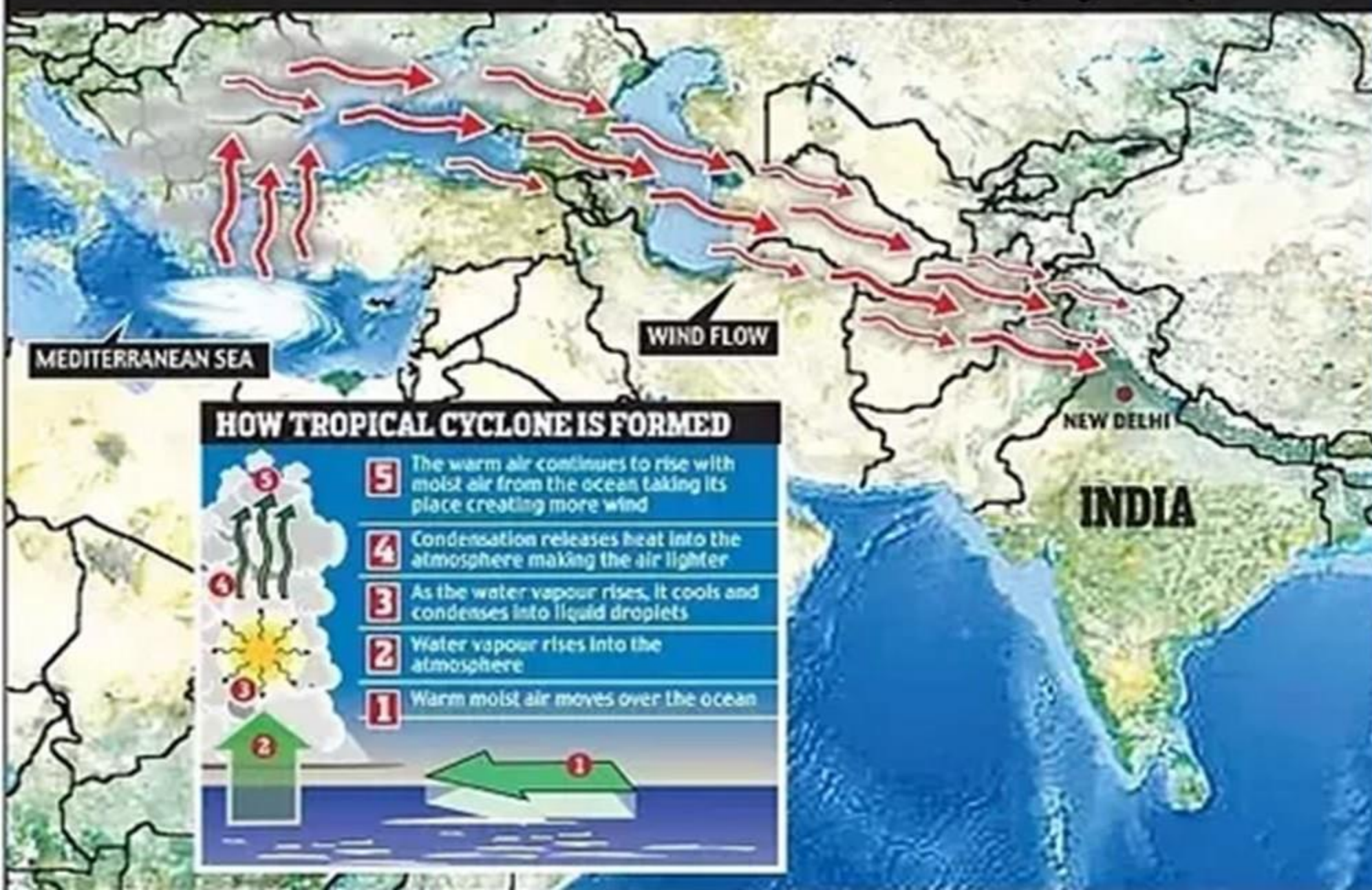
WESTERN DISTURBANCE

■ Western disturbance is a cyclonic circulation in the mid and lower troposphere (between 2.1 and 7.6 km above sea level) which originates over the Mediterranean Sea, Caspian Sea and Black Sea

■ It is one of the most important weather systems that causes adverse weather conditions over North-West India and particularly over the western Himalayan region

■ Western disturbance during winter leads to occurrence of clouds, rainfall and snowfall

■ An intense western disturbance usually produces widespread heavy snowfall over the western Himalayan region and rains over northern plains for a day or two and may trigger avalanches



HOW TROPICAL CYCLONE IS FORMED

- 1 Warm moist air moves over the ocean
- 2 Water vapour rises into the atmosphere
- 3 As the water vapour rises, it cools and condenses into liquid droplets
- 4 Condensation releases heat into the atmosphere making the air lighter
- 5 The warm air continues to rise with moist air from the ocean taking its place creating more wind

FACTORS RELATED TO AIR PRESSURE & WINDS

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MECHANISM OF WEATHER IN THE WINTER SEASON

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WESTERN CYCLONIC DISTURBANCE AND TROPICAL CYCLONES

WHY WAS IT IN NEWS

- **INDIAN EXPRESS:**
Frequent western disturbances behind thunderstorms in north India

WHAT ARE WESTERN DISTURBANCES?

- Because of the high terrain, mountain ranges, most of these disturbances are in the mature stage and hence weak or irregular by the time they reach India.
- Reaching Rajasthan, Haryana and Punjab, these disturbances slow down and stagnate due to the nearly closed in nature of the region with high hills.
- The tracks of these disturbances come farthest south to latitudes $22/23^{\circ}\text{N}$ in the month of February.
- An increase in the prevailing night temperature generally indicates an advance in the arrival of these disturbances.

FACTORS RELATED TO AIR PRESSURE & WINDS

1 MECHANISM OF WEATHER IN THE WINTER SEASON

FACTOR NO:3

WESTERN CYCLONIC DISTURBANCE AND TROPICAL CYCLONES

The western disturbances result in:

- Winter rainfall in Punjab, Haryana, Delhi and western Uttar Pradesh, locally known as 'mahawat'. Although the amount is meagre, it is highly beneficial for Rabi crops.
- Snowfall in the higher altitudes of Kashmir and Himachal Pradesh and sometimes hail too. It is this snow that sustains the flow of water in the Himalayan rivers during the summer months.
- The precipitation in this period goes on decreasing from west to east in the plains and from north to south in the mountains.

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